Name
Date
Course

Beginning Algebra Practice Test II (version 1)

Instructions

i) As always, write a title, name, date, and course.
ii) Number each problem, circle the number, and show work when necessary.
iii) Draw a rectangle around your answers.
iv) Staple a copy of these test questions with your work and your final answers.

Simplify.

1) $$\frac{x^2-4}{x^2-6x+8} = \frac{(x+2)(x-2)}{(x-4)(x-2)} = \boxed{\frac{x+2}{x-4}}$$

2) $$\frac{3x+2}{9x^2-4} = \frac{3x+2}{(3x-2)(3x+2)} = \boxed{\frac{1}{3x-2}}$$

3) $$\frac{8x^5-16x^4-10x^3}{5x^2-2x^3} = \frac{2x^3(4x^2-8x-5)}{x^2(5-2x)} = \frac{2x^3(2x+1)(2x-5)}{x^2(5-2x)}$$

$$= \frac{2x^3(2x+1)(2x-5)}{-x^2(-5+2x)}$$

$$= \boxed{-2x(2x+1)}$$

Perform the indicated operations.

4) $$\frac{x^2-1}{14y^2-9xy+x^2} \cdot \frac{7y-x}{x-1}$$

$$\frac{(x-1)(x+1)}{(7y-x)(2y-x)} \cdot \frac{7y-x}{x-1}$$

$$\boxed{\frac{x+1}{2y-x}}$$

5) $$\frac{x^2+7x}{x^2+5x-14} \div \frac{5x^5+35x^4}{x+1}$$

$$\frac{x^2+7x}{x^2+5x-14} \cdot \frac{x+1}{5x^5+35x^4}$$

$$\frac{x(x+7)}{(x+7)(x-2)} \cdot \frac{x+1}{5x^4(x+7)}$$

$$\boxed{\frac{x+1}{5x^3(x-2)(x+7)}}$$

6) $$\frac{3x}{x^2+3x+2} + \frac{5}{x+1}$$

$$\frac{3x}{(x+1)(x+2)} + \frac{5}{x+1}$$

$$\frac{3x}{(x+1)(x+2)} + \frac{5(x+2)}{(x+1)(x+2)}$$

$$\frac{3x}{(x+1)(x+2)} + \frac{5x+10}{(x+1)(x+2)}$$

$$\boxed{\frac{8x+10}{(x+1)(x+2)}}$$

7) $$\frac{2}{3x} - \frac{x+7}{x(x-3)}$$

$$\frac{2(x-3)}{3x(x-3)} - \frac{3(x+7)}{3x(x-3)}$$

$$\frac{2x-6}{3x(x-3)} - \frac{3x+21}{3x(x-3)}$$

$$\frac{2x-6-3x-21}{2x(x-3)}$$

$$\boxed{\frac{-x-27}{2x(x-3)}}$$

Find the square roots. You will have to approximate one of the roots with a calculator (Round to the nearest thousandth).

(8) a) $\sqrt{\frac{36}{81}} = \frac{\sqrt{36}}{\sqrt{81}} = \frac{6}{9} = \boxed{\frac{2}{3}}$ b) $\sqrt{16} = \boxed{4}$

c) $\sqrt{17} \approx \boxed{4.123}$ d) $\sqrt{1} = \boxed{1}$

(9) Factor out a perfect square:

a) $\sqrt{20} = \sqrt{4 \cdot 5} = \sqrt{4} \cdot \sqrt{5} = 2 \cdot \sqrt{5} = \boxed{2\sqrt{5}}$

b) $\sqrt{45} = \sqrt{9 \cdot 5} = \sqrt{9} \cdot \sqrt{5} = 3 \cdot \sqrt{5} = \boxed{3\sqrt{5}}$

Add or subtract

c) $\sqrt{7} + 4\sqrt{7} - 10\sqrt{7}$

$5\sqrt{7} - 10\sqrt{7}$

$\boxed{-5\sqrt{7}}$

d) $-8\sqrt{3} - 2\sqrt{3} + 4\sqrt{3}$

$-10\sqrt{3} + 4\sqrt{3}$

$\boxed{-6\sqrt{3}}$

Multiply. Never leave perfect square factors under the root signs.

(10) a) $\sqrt{2} \cdot \sqrt{2} = \sqrt{2 \cdot 2} = \sqrt{4} = \boxed{2}$

b) $-\sqrt{6} \cdot 3\sqrt{10} = -3\sqrt{6 \cdot 10} = -3\sqrt{60} = -3\sqrt{4 \cdot 15}$

$= -3\sqrt{4} \cdot \sqrt{15}$

$= -3 \cdot 2 \cdot \sqrt{15}$

$= -6 \cdot \sqrt{15}$

$= \boxed{-6\sqrt{15}}$

c) $4\sqrt{14}(-3\sqrt{2})$

$-12\sqrt{14 \cdot 2}$

$-12\sqrt{28}$

$-12\sqrt{4 \cdot 7}$

$-12\sqrt{4} \cdot \sqrt{7}$

$-12 \cdot 2 \cdot \sqrt{7}$

$-24 \cdot \sqrt{7}$

$\boxed{-24\sqrt{7}}$

d) $-9\sqrt{5}(2\sqrt{2} - 6\sqrt{5} + \sqrt{15})$

$-18\sqrt{5 \cdot 2} + 54\sqrt{5 \cdot 5} - 9\sqrt{5 \cdot 15}$

$-18\sqrt{10} + 54\sqrt{25} - 9\sqrt{75}$

$-18\sqrt{10} + 54 \cdot 5 - 9\sqrt{25 \cdot 3}$

$-18\sqrt{10} + 270 - 9\sqrt{25} \cdot \sqrt{3}$

$-18\sqrt{10} + 270 - 9 \cdot 5 \cdot \sqrt{3}$

$\boxed{-18\sqrt{10} + 270 - 45\sqrt{3}}$

(11) Divide or multiply.

a) $-8(-9\sqrt{11}) = \boxed{72\sqrt{11}}$

b) $\frac{\sqrt{72}}{\sqrt{8}} = \sqrt{\frac{72}{8}} = \sqrt{9} = \boxed{3}$

Rationalize the denominators.

c) $\frac{2}{\sqrt{6}}$

$$\frac{2\cdot\sqrt{6}}{\sqrt{6}\cdot\sqrt{6}}$$

$$\frac{2\sqrt{6}}{\sqrt{6\cdot 6}}$$

$$\frac{2\sqrt{6}}{\sqrt{36}}$$

$$\frac{2\sqrt{6}}{6}$$

$$\boxed{\frac{\sqrt{6}}{3}}$$

d) $\frac{12}{\sqrt{32}}$

$$\frac{12}{\sqrt{16\cdot 2}}$$

$$\frac{12}{\sqrt{16}\cdot\sqrt{2}}$$

$$\frac{12}{4\cdot\sqrt{2}}$$

$$\frac{3}{\sqrt{2}}$$

$$\frac{3\cdot\sqrt{2}}{\sqrt{2}\cdot\sqrt{2}}$$

$$\frac{3\sqrt{2}}{\sqrt{2\cdot 2}}$$

$$\frac{3\sqrt{2}}{\sqrt{4}} = \boxed{\frac{3\sqrt{2}}{2}}$$

Simplify.

(12) a) $\dfrac{\frac{2}{3}}{\frac{4}{5}} = \frac{2}{3} \div \frac{4}{5} = \frac{2}{3}\cdot\frac{5}{4} = \frac{10}{12} = \boxed{\frac{5}{6}}$

b) $\dfrac{4}{\frac{1}{3}} = 4 \div \frac{1}{3} = \frac{4}{1} \div \frac{1}{3} = \frac{4}{1}\cdot\frac{3}{1} = \frac{12}{1} = \boxed{12}$

c) $\dfrac{\frac{8}{7}}{-2x} = \frac{8}{7} \div (-2x) = \frac{8}{7} \div \left(-\frac{2x}{1}\right) = \frac{8}{7}\cdot\left(-\frac{1}{2x}\right) = -\frac{8}{14x}$

$= \boxed{-\frac{4}{7x}}$

d) $\dfrac{\frac{3}{8}-\frac{1}{5}}{\frac{7}{20}+\frac{9}{10}} = \dfrac{\frac{15}{40}-\frac{8}{40}}{\frac{7}{20}+\frac{18}{20}} = \dfrac{\frac{7}{40}}{\frac{25}{20}} = \dfrac{\frac{7}{40}}{\frac{5}{4}} = \frac{7}{40} \div \frac{5}{4}$

$= \frac{7}{40}\cdot\frac{4}{5}$

$= \frac{28}{200}$

$= \boxed{\frac{7}{50}}$

Divide.

(13) $\dfrac{-30x^3-25x}{2-x^2} \div \dfrac{6x^4+17x^2+10}{x^4-4}$

$\dfrac{-30x^3-25x}{2-x^2} \cdot \dfrac{x^4-4}{6x^4+17x^2+10}$

$\dfrac{-5x(6x^2+5)}{2-x^2} \cdot \dfrac{(x^2+2)(x^2-2)}{(6x^2+5)(x^2+2)}$

$\dfrac{-5x(6x^2+5)}{-1(-2+x^2)} \cdot \dfrac{(x^2+2)(x^2-2)}{(6x^2+5)(x^2+2)}$

$\dfrac{-5x}{-1} = \boxed{5x}$

GCF
Quadratic
DOTS
Grouping
Multivariate
ax^4+bx^2+c

Simplify (i.e. add and/or subtract).

(14) a) $-10\sqrt{7} - \sqrt{40} + 3\sqrt{10} + 3\sqrt{63}$

$-10\sqrt{7} - \sqrt{4\cdot10} + 3\sqrt{10} + 3\sqrt{9\cdot7}$

$-10\sqrt{7} - \sqrt{4}\cdot\sqrt{10} + 3\sqrt{10} + 3\sqrt{9}\cdot\sqrt{7}$

$-10\sqrt{7} - 2\sqrt{10} + 3\sqrt{10} + 3\cdot3\sqrt{7}$

$-10\sqrt{7} - 2\sqrt{10} + 3\sqrt{10} + 9\sqrt{7}$

$\boxed{-\sqrt{7} + \sqrt{10}}$

Factor out a perfect square(s) (leave no perfect square factor in the root).

b) $\sqrt{180} = \sqrt{18\cdot10} = \sqrt{9\cdot2\cdot5\cdot2} = \sqrt{9\cdot4\cdot5} = \sqrt{36\cdot5}$

$= \sqrt{36}\cdot\sqrt{5}$

$= \boxed{6\sqrt{5}}$

Rationalize the denominators.

(15) a) $\sqrt{\dfrac{1}{64}}$ b) $\sqrt{\dfrac{3}{10}}$ c) $\sqrt{\dfrac{49}{5}}$ d) $\sqrt{\dfrac{13}{36}}$

a) $\dfrac{\sqrt{1}}{\sqrt{64}}$

$\boxed{\dfrac{1}{8}}$

b) $\dfrac{\sqrt{3}}{\sqrt{10}}$

$\dfrac{\sqrt{3}\cdot\sqrt{10}}{\sqrt{10}\cdot\sqrt{10}}$

$\dfrac{\sqrt{3\cdot10}}{\sqrt{10\cdot10}}$

$\dfrac{\sqrt{30}}{\sqrt{100}}$

$\boxed{\dfrac{\sqrt{30}}{10}}$

c) $\dfrac{\sqrt{49}}{\sqrt{5}}$

$\dfrac{7}{\sqrt{5}}$

$\dfrac{7\cdot\sqrt{5}}{\sqrt{5}\cdot\sqrt{5}}$

$\dfrac{7\sqrt{5}}{\sqrt{5\cdot5}}$

$\dfrac{7\sqrt{5}}{\sqrt{25}}$

$\boxed{\dfrac{7\sqrt{5}}{5}}$

d) $\dfrac{\sqrt{13}}{\sqrt{36}}$

$\boxed{\dfrac{\sqrt{13}}{6}}$

Simplify.

(16) a) $\dfrac{\frac{5x}{6}}{\frac{-1}{x+4}} = \dfrac{5x}{6} \div \dfrac{-1}{x+4} = \dfrac{5x}{6} \cdot \dfrac{x+4}{-1} = \dfrac{5x(x+4)}{-6}$

$= \boxed{\dfrac{5x^2+20x}{-6}}$

Rationalize the denominator.

b) $\sqrt{\frac{11}{50}} = \dfrac{\sqrt{11}}{\sqrt{50}} = \dfrac{\sqrt{11}}{\sqrt{25\cdot 2}} = \dfrac{\sqrt{11}}{\sqrt{25}\cdot\sqrt{2}} = \dfrac{\sqrt{11}}{5\sqrt{2}}$

$= \dfrac{\sqrt{11}\cdot\sqrt{2}}{5\sqrt{2}\cdot\sqrt{2}}$

$= \dfrac{\sqrt{11\cdot 2}}{5\sqrt{2\cdot 2}}$

$= \dfrac{\sqrt{22}}{5\sqrt{4}}$

$= \dfrac{\sqrt{22}}{5\cdot 2}$

$= \boxed{\dfrac{\sqrt{22}}{10}}$

Simplify.

(17) $\dfrac{\frac{2x^3}{5y^4}}{\frac{2z}{9yx}} = \dfrac{2x^3}{5y^4} \div \dfrac{2z}{9yx} = \dfrac{2x^3}{5y^4} \cdot \dfrac{9yx}{2z}$

$= \dfrac{18x^4y}{10y^4z}$

$= \boxed{\dfrac{9x^4}{5y^3z}}$

(18) Add and/or subtract.

$-\dfrac{2}{b-6} + \dfrac{3}{4b} - \dfrac{2}{b}$

$-\dfrac{2(4b)}{(b-6)(4b)} + \dfrac{3(b-6)}{4b(b-6)} - \dfrac{2\cdot 4(b-6)}{b\cdot 4(b-6)}$

$-\dfrac{8b}{4b(b-6)} + \dfrac{3b-18}{4b(b-6)} - \dfrac{8b-48}{4b(b-6)}$

$\dfrac{-8b+3b-18-8b+48}{4b(b-6)}$

$\boxed{\dfrac{-13b+30}{4b(b-6)}}$

Rationalize the denominators.

$$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$$

$$\sqrt{a}\cdot\sqrt{b} = \sqrt{ab}$$

(19) a) $\sqrt{\frac{48}{5}}$

$$\frac{\sqrt{48}}{\sqrt{5}}$$

$$\frac{\sqrt{16\cdot 3}}{\sqrt{5}}$$

$$\frac{\sqrt{16}\cdot\sqrt{3}}{\sqrt{5}}$$

$$\frac{4\sqrt{3}}{\sqrt{5}}$$

$$\frac{4\sqrt{3}\cdot\sqrt{5}}{\sqrt{5}\cdot\sqrt{5}}$$

$$\frac{4\sqrt{3\cdot 5}}{\sqrt{5\cdot 5}}$$

$$\frac{4\sqrt{15}}{\sqrt{25}} = \boxed{\frac{4\sqrt{15}}{5}}$$

b) $\sqrt{\frac{56}{27}}$

$$\frac{\sqrt{56}}{\sqrt{27}}$$

$$\frac{\sqrt{4\cdot 14}}{\sqrt{9\cdot 3}}$$

$$\frac{\sqrt{4}\,\sqrt{14}}{\sqrt{9}\cdot\sqrt{3}}$$

$$\frac{2\cdot\sqrt{14}}{3\cdot\sqrt{3}}$$

$$\frac{2\sqrt{14}}{3\sqrt{3}}$$

$$\frac{2\sqrt{14}\cdot\sqrt{3}}{3\sqrt{3}\cdot\sqrt{3}}$$

$$\frac{2\sqrt{14\cdot 3}}{3\sqrt{3\cdot 3}} = \frac{2\sqrt{42}}{3\sqrt{9}}$$

$$= \frac{2\sqrt{42}}{3\cdot 3}$$

$$= \boxed{\frac{2\sqrt{42}}{9}}$$

Add.

(20) $\frac{-7}{x^2-3x-18} + \frac{2}{x^2+6x+9}$

$$\frac{-7}{(x-6)(x+3)} + \frac{2}{(x+3)(x+3)}$$

$$\frac{-7(x+3)}{(x-6)(x+3)(x+3)} + \frac{2(x-6)}{(x+3)(x+3)(x-6)}$$

$$\frac{-7x-21}{(x-6)(x+3)(x+3)} + \frac{2x-12}{(x+3)(x+3)(x-6)}$$

$$\boxed{\frac{-5x-33}{(x-6)(x+3)(x+3)}}$$

Add and/or Subtract.

(21) $\frac{5}{x-10} + \frac{9}{x^2-100} - \frac{x-2}{x+10}$

$$\frac{5}{x-10} + \frac{9}{(x-10)(x+10)} - \frac{x-2}{x+10}$$

$$\frac{5(x+10)}{(x-10)(x+10)} + \frac{9}{(x-10)(x+10)} - \frac{(x-2)(x-10)}{(x+10)(x-10)}$$

$$\frac{5x+50}{(x-10)(x+10)} + \frac{9}{(x-10)(x+10)} - \frac{x^2-12x+20}{(x+10)(x-10)}$$

$$\frac{5x+50+9-x^2+12x-20}{(x-10)(x+10)}$$

$$\boxed{\frac{-x^2+17x+39}{(x-10)(x+10)}}$$

Add and/or Subtract.

(22) $\dfrac{x}{4x-1}+\dfrac{8}{4x^2+19x-5}-\dfrac{4x^2}{x+5}$

$$\frac{x}{4x-1}+\frac{8}{(4x-1)(x+5)}-\frac{4x^2}{x+5}$$

$$\frac{x(x+5)}{(4x-1)(x+5)}+\frac{8}{(4x-1)(x+5)}-\frac{4x^2(4x-1)}{(x+5)(4x-1)}$$

$$\frac{x^2+5x}{(4x-1)(x+5)}+\frac{8}{(4x-1)(x+5)}-\frac{16x^3-4x^2}{(x+5)(4x-1)}$$

$$\frac{x^2+5x+8-16x^3+4x^2}{(4x-1)(x+5)}$$

$$\boxed{\frac{-16x^3+5x^2+5x+8}{(4x-1)(x+5)}}$$

Simplify.

(23) $\dfrac{\dfrac{-2}{3x^3+21x^2}}{\dfrac{1}{x^2+8x+7}}=\dfrac{-2}{3x^3+21x^2}\div\dfrac{1}{x^2+8x+7}$

$$=\frac{-2}{3x^3+21x^2}\cdot\frac{x^2+8x+7}{1}$$

$$=\frac{-2}{3x^2(x+7)}\cdot\frac{(x+7)(x+1)}{1}$$

$$=\boxed{\frac{-2(x+1)}{3x^2}}$$

(24) $$\frac{\frac{1}{x+3}-\frac{6x+2}{x^2-2x-15}}{\frac{4}{x-5}+1}=\frac{\frac{1}{x+3}-\frac{6x+2}{(x-5)(x+3)}}{\frac{4}{x-5}+\frac{1}{1}}$$

$$=\frac{\frac{1(x-5)}{(x+3)(x-5)}-\frac{6x+2}{(x-5)(x+3)}}{\frac{4}{x-5}+\frac{1(x-5)}{1(x-5)}}$$

$$=\frac{\frac{x-5}{(x+3)(x-5)}-\frac{6x+2}{(x-5)(x+3)}}{\frac{4}{x-5}+\frac{x-5}{x-5}}$$

$$=\frac{\frac{x-5-6x-2}{(x+3)(x-5)}}{\frac{4+x-5}{x-5}}$$

$$=\frac{\frac{-5x-7}{(x+3)(x-5)}}{\frac{-1+x}{x-5}}$$

$$=\frac{-5x-7}{(x+3)(x-5)}\div\frac{-1+x}{x-5}$$

$$=\frac{-5x-7}{(x+3)(x-5)}\cdot\frac{x-5}{-1+x}$$

$$=\boxed{\frac{-5x-7}{(x+3)(-1+x)}}$$

Add and/or subtract.

(25) $$3-\frac{-4+x}{25x^2-64}+\frac{1}{8-5x}$$

$$\frac{3}{1}-\frac{-4+x}{(5x+8)(5x-8)}+\frac{1}{-1(-8+5x)}$$

$$\frac{3(5x+8)(5x-8)}{1(5x+8)(5x-8)}-\frac{-4+x}{(5x+8)(5x-8)}+\frac{-1}{-8+5x}$$

$$\frac{3(25x^2-64)}{1(25x^2-64)}-\frac{-4+x}{(5x+8)(5x-8)}+\frac{-1(5x+8)}{(-8+5x)(5x+8)}$$

$$\frac{75x^2-192}{25x^2-64}-\frac{-4+x}{(5x+8)(5x-8)}+\frac{-5x-8}{(-8+5x)(5x+8)}$$

$$\frac{75x^2-192+4-x-5x-8}{25x^2-64}$$

$$\boxed{\frac{75x^2-6x-196}{25x^2-64}}$$